

Yutong (Kelly) He

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Education

- 2022 – 2028
(Expected) **Carnegie Mellon University** – Pittsburgh, PA
PhD in Machine Learning, School of Computer Science
Advisors: Zico Kolter, Ruslan Salakhutdinov
- 2019 – 2022 **Stanford University** – Stanford, CA
MS in Computer Science (with Distinction in Research)
Advisors: Stefano Ermon
GPA: 4.20/4.30
- 2015 – 2019 **University of Rochester** – Rochester, NY
BS in Mathematics (with Highest Distinction)
BS in Data Science (with Highest Distinction)
GPA: 3.95/4.00

Honors and Scholarships

- 2026 Qualcomm Innovation Fellowship
- 2025 NeurIPS 2025 Top Reviewer
- 2022 Outstanding Paper at ICLR 2022
- 2021 2nd place in Alexa Prize Socialbot Grand Challenge 4
- 2020 Siebel Scholar 2020-2021
- 2019-2020 Best Project Awards at Stanford CS 229, CS 224N, CS 230
- 2019 Doris Ermine Smith Award for Achievement in Mathematics
- 2019 Phi Beta Kappa
- 2018 University of Rochester Research Presentation Award
- 2018 University of Rochester Discovery Grant
- 2017 Xerox Engineering Research Fellowship

Publications

- 2026 **Diamond Maps: Efficient Reward Alignment via Stochastic Flow Maps**
Peter Holderrieth*, Douglas Chen*, Luca Eyring*, Ishin Shah, Giri Anantharaman, **Yutong He**, Zeynep Akata, Tommi Jaakkola, Nicholas Matthew Boffi, Max Simchowitz
International Conference on Machine Learning (ICML 2026)
- 2026 **Joint Distillation for Fast Likelihood Evaluation and Sampling in Flow-based Models**
Xinyue Ai*, **Yutong He***, Albert Gu, Ruslan Salakhutdinov, J. Zico Kolter, Nicholas M. Boffi, Max Simchowitz
International Conference on Learning Representations (ICLR 2026)
- 2026 **Accelerating Diffusion Models in Offline RL via Reward-Aware Consistency Trajectory Distillation**
Xintong Duan*, **Yutong He***, Fahim Tajwar, Russ Salakhutdinov, J Zico Kolter, Jeff Schneider
International Conference on Learning Representations (ICLR 2026)

- 2025 **Automated Black-box Prompt Engineering for Personalized Text-to-Image Generation**
Yutong He, Alexander Robey, Naoki Murata, Yiding Jiang, Joshua Williams, George J. Pappas, Hamed Hassani, Yuki Mitsufuji, Ruslan Salakhutdinov, J. Zico Kolter
Transactions on Machine Learning Research (TMLR 2025)
- 2025 **Blind Inverse Problem Solving Made Easy by Text-to-Image Latent Diffusion**
Michail Dontas*, Yutong He*, Naoki Murata, Yuki Mitsufuji, J Zico Kolter, Ruslan Salakhutdinov
Neural Information Processing Systems (NeurIPS 2025) workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)
- 2025 **State Combinatorial Generalization In Decision Making With Conditional Diffusion Models**
Xintong Duan, Yutong He, Fahim Tajwar, Wentse Chen, Russ Salakhutdinov, Jeff Schneider
Transactions on Machine Learning Research (TMLR 2025)
- 2025 **Prompt Recovery for Image Generation Models: A Comparative Study of Discrete Optimizers**
Joshua Nathaniel Williams, Avi Schwarzschild, Yutong He, J. Zico Kolter
arXiv:2408.06502
- 2024 **Weather Prediction with Diffusion Guided by Realistic Forecast Processes**
Zhanxiang Hua*, Yutong He*, Chengqian Ma, Alexandra Anderson-Frey
arXiv:2402.06666
- 2024 **Manifold Preserving Guided Diffusion**
Yutong He*, Naoki Murata*, Chieh-Hsin Lai, Yuhta Takida, Toshimitsu Uesaka, Dongjun Kim, Wei-Hsiang Liao, Yuki Mitsufuji, J. Zico Kolter, Ruslan Salakhutdinov, Stefano Ermon
International Conference on Learning Representations (ICLR 2024)
- 2024 **Consistency Trajectory Models: Learning Probability Flow ODE Trajectory of Diffusion**
Dongjun Kim, Chieh-Hsin Lai, Wei-Hsiang Liao, Naoki Murata, Yuhta Takida, Toshimitsu Uesaka, Yutong He, Yuki Mitsufuji, Stefano Ermon
International Conference on Learning Representations (ICLR 2024)
- 2023 **Towards reporting bias in visual-language datasets: bimodal augmentation by decoupling object-attribute association**
Qiyu Wu, Mengjie Zhao, Yutong He, Lang Huang, Junya Ono, Hiromi Wakaki, Yuki Mitsufuji
arXiv:2310.01330
- 2023 **Localized Text-to-Image Generation for Free via Cross Attention Control**
Yutong He, Ruslan Salakhutdinov, J. Zico Kolter
arXiv:2306.14636
- 2023 **CSP: Self-Supervised Contrastive Spatial Pre-Training for Geospatial-Visual Representations**
Gengchen Mai, Ni Lao, Yutong He, Jiaming Song, Stefano Ermon
International Conference on Machine Learning (ICML 2023)
- 2022 **SatMAE: Pre-training Transformers for Temporal and Multi-Spectral Satellite Imagery**
Yezhen Cong*, Samar Khanna*, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David B. Lobell, Stefano Ermon
Neural Information Processing Systems (NeurIPS 2022)

- 2022 **Understanding Economic Development in Rural Africa using Satellite Imagery, Building footprints and Deep Models**
Amna Elmustafa, Erik Rozi, **Yutong He**, Gengchen Mai, Stefano Ermon, Marshall Burke, David Lobell
International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL 2022)
- 2022 **Neural Generation Meets Real People: Building a Social, Informative Open-Domain Dialogue Agent**
Ethan A. Chi, Caleb Chiam, Trenton Chang, Swee Kiat Lim, Chetanya Rastogi, Alexander Iyabor, **Yutong He**, Hari Sowrirajan, Avanika Narayan, Jillian Tang, Haojun Li, Ashwin Paranjape, Christopher D. Manning
The 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL 2022)
- 2022 **Comparing Distributions by Measuring Differences that Affect Decision Making**
Shengjia Zhao*, Abhishek Sinha*, **Yutong He***, Aidan Perreault, Jiaming Song, Stefano Ermon
International Conference on Learning Representations (ICLR 2022)
Outstanding Paper Award (Top 7/4492)
- 2022 **SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations**
Chenlin Meng, **Yutong He**, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, Stefano Ermon
International Conference on Learning Representations (ICLR 2022)
- 2021 **Spatial-Temporal Super-Resolution of Satellite Imagery via Conditional Pixel Synthesis**
Yutong He, Dingjie Wang, Nicholas Lai, William Zhang, Chenlin Meng, Marshall Burke, David B. Lobell, Stefano Ermon
Neural Information Processing Systems (NeurIPS 2021)
- 2021 **Tracking Urbanization in Developing Regions with Remote Sensing Spatial-Temporal Super-Resolution**
Yutong He*, William Zhang*, Chenlin Meng, Marshall Burke, David B. Lobell, Stefano Ermon
Neural Information Processing Systems (NeurIPS 2021) workshop on Machine Learning for the Developing World (ML4D)
- 2021 **Neural, Neural Everywhere: Controlled Generation Meets Scaffolded, Structured Dialogue**
Ethan A. Chi, Caleb Chiam, Trenton Chang, Swee Kiat Lim, Chetanya Rastogi, Alexander Iyabor, **Yutong He**, Hari Sowrirajan, Avanika Narayan, Jillian Tang, Haojun Li, Ashwin Paranjape, Christopher D. Manning
Alexa Prize Proceedings 2021
- 2020 **Fine-grained Image-to-Image Transformation towards Visual Recognition**
Wei Xiong, **Yutong He**, Yixuan Zhang, Wenhan Luo, Lin Ma, and Jiebo Luo
International Conference on Computer Vision and Pattern Recognition (CVPR 2020)
- 2020 **Motion-based Handwriting Recognition and Word Reconstruction**
Junshen Kevin Chen*, Wanze Xie*, **Yutong He***
arXiv:2101.06025

Industry Experience

- Summer 2026 **Google UK Limited (Student Researcher Intern)** – London, UK
Working in Google DeepMind on diffusion models.
- Summer 2025 **Meta Platforms, Inc. (Research Scientist Intern)** – New York, NY
Worked in FAIR on flow-based models.

- Summer 2023 **Sony Group Corporation (Student Research Scientist Intern)** – Tokyo, Japan
Worked in Sony Creative AI team on controllable image generation.
- Summer 2020 **Adobe Inc. (Machine Learning Engineer Intern)** – San Jose, CA
Worked in Sensei & Search team on visual-textual search and recommendation for E-commerce.

Teaching Experience

- Spring 2026 **10799: Diffusion & Flow Matching (Carnegie Mellon University)**
Instructor who designed the entire class from scratch
- Fall 2025 **10417: Intermediate Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Spring 2025 **10708: Probabilistic Graphical Models (Carnegie Mellon University)**
TA and lecturer on Markov chain Monte Carlo and diffusion models
- Spring 2025 **10707: Advanced Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Fall 2024 **10708: Probabilistic Graphical Models (Carnegie Mellon University)**
TA and lecturer on Markov chain Monte Carlo and diffusion models
- Fall 2024 **10417: Intermediate Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Spring 2024 **10707: Advanced Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Fall 2023 **10417: Intermediate Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Spring 2023 **10707: Advanced Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Fall 2022 **10417: Intermediate Deep Learning (Carnegie Mellon University)**
Guest lecturer on Diffusion Models
- Winter 2022 **CS 228: Probabilistic Graphical Models (Stanford)**
TA and lecturer on Markov chain Monte Carlo
- Fall 2021 **CS 236: Deep Generative Models (Stanford)**
TA and lecturer on neural networks
- Winter 2021 **CS 228: Probabilistic Graphical Models (Stanford)**
TA and lecturer on Markov chain Monte Carlo
- Spring 2019 **CSC 249/449: Machine Vision (University of Rochester)**
Teaching Assistant
- Fall 2018 **DSC 262/462: Computational Introduction to Statistics (University of Rochester)**
Teaching Assistant
- Spring 2018 **CSC 242/442: Data Mining (University of Rochester)**
Teaching Assistant
- Spring 2018 **MTH 150: Discrete Mathematics (University of Rochester)**
Teaching Assistant
- Fall 2017 **CSC 242/442: Artificial Intelligence (University of Rochester)**
Teaching Assistant
- Fall 2017 **CSC 261/461: Database System (University of Rochester)**
Teaching Assistant
- Fall 2017 **MTH 201: Introduction to Probability (University of Rochester)**
Teaching Assistant

Spring 2017	CSC 172: Data Structures and Algorithms (University of Rochester) Workshop Leader
Spring 2017	MTH 162: Calculus IIA (University of Rochester) Workshop Leader
Fall 2016	CSC 171: Introduction to Computer Science (University of Rochester) Workshop Leader
Fall 2016	MTH 141: Calculus I (University of Rochester) Workshop Leader

Leadership and Mentorship

2021	Stanford AI4ALL (Computer Vision Mentor) Led a group of high school students from under-represented populations to complete a hands-on research project in computer vision, and provided them exposure to a variety of AI topics, in-depth discussions of cutting-edge AI research, and exploration of the humanistic and societal impact of AI.
2021	Stanford CURIS Program for Undergraduate Research (Mentor) Mentored a group of undergraduate students on artificial intelligence research projects.
2020	Stanford Summer Undergraduate Research Fellowship (Graduate Student Mentor) Advised Summer Undergraduate Research Fellowship (SURF) scholars to reflect on their summer research/professional development experience, learn about the graduate school application process, and gain insight into graduate student life, specifically at Stanford.
2018-2019	University of Rochester Computer Science Undergraduate Council (President) • Served as a representative of the undergraduate students in computer science community in University of Rochester and a bridge of communication between computer science undergraduate students and the graduate students, faculty members, other departments and other schools. • Hosted university hackathons, department town halls, social events, and panels. • Organized teams to attend international programming competitions.
2018-2019	University of Rochester Goergen Institute for Data Science (Peer Advisor) Advised students on declaring majors, making connections with faculty members, reviewing research opportunities, and exploring interdisciplinary study.
2016-2019	University of Rochester Computer Science Undergraduate Council (Tutor) Held weekly voluntary tutoring session for computer science, mathematics, statistics courses.